Grid Connected Hybrid Power System Design Using HOMER

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***Abstract*—This paper presents an economical expediency of grid connected hybrid (PV/Wind turbine) power system for rural area applications in the southern city of Bangladesh, Lobon Chora, Khulna. The analysis is carried out by investigating the potentials of wind and solar energy and collecting data from different sources. Hybrid Optimization Model for Electric Renewable (HOMER) software is used to analyze the available data and economic feasibility of the proposed hybrid power system. In this paper, two types of models named off-grid and on-grid are designed and optimized for comparison purposes. Also, a sensitivity analysis has been performed for each model in order to determine the effect of variations in grid energy cost on its total cost. The simulation results indicate that the proposed grid connected hybrid (PV/Wind turbine) power system is most suitable and cost competitive for the mentioned region.**

Keywords—Grid; HOMER; hybrid power system; renewable energy.

I. INTRODUCTION

Renewable energy sources (RES) offer an efficient solution to the global warming and high fuel cost. Therefore, there is a growing interest in RES, particularly photovoltaic (PV) and wind energy [1], [2]. Recently, hybrid power systems using RE are becoming most popular due to their potential advantages [3].

Although wind and solar has several advantages, a solar or wind generator in a stand-alone system cannot supply the load continuously due to their intermittent nature [4]. As load demands are always changing with time, the changes in solar or wind energy generations do not always match with the time distribution of consumer's demand [5]. Therefore, there is a need of additional battery storage or other components for providing continuous power supply to the load. It has been investigated that a hybrid PV/Wind/battery system is a secure source of electricity [6], [7], [8]. However, due to the high cost of battery energy storage, a stand-alone system is very expensive [9]. Therefore, it is essential to find an effective solution to this problem. On the other hand, a grid connected hybrid power system with proper planning requires storage with less capacity which, in turn, reduces the cost.

Therefore, the main goal of this paper is to analyze the economic feasibility of grid connected hybrid (PV/Wind turbine) power system to meet the load requirement on a specific area. For a comparison purpose, an off-grid hybrid (PV/Wind/battery) power system is also designed. The rest of the paper is organized as follows. Section II describes the system model. Methodology is given in Section III. Simulation models are provided in Section IV and optimization results are evaluated in Section V. Finally, conclusions are drawn in Section VI.

II. SYSTEM DESCRIPTION

In this paper, both on and off-grid hybrid power system models have been designed using HOMER to evaluate and determine the cost of different models. HOMER simulation software needs some input data to evaluate the optimization results for different combinations which are described in the next section [4].

A. Load Profile

The considered area is dominated by irrigation pumps during peak demand. In this paper, the load profile considered average energy consumption by the proposed area and it is 654.73 kWh/d. Figure 1 shows the daily average load profile in which 17:00 to 23:00 h time period has been found as peak demand which determines the size of the system. Here, 101.32 kW is considered as peak load consumption. The scaled annual average (kWh/d) is assumed 654.73. Fig. 2 shows the monthly average load in the proposed region.

B. Wind Speed and Solar Radiation

The data of wind speed and solar radiation is obtained from NASA surface meteorology and solar energy database. Wind speed data obtained at 50 m above the surface of sea level for the location of Khulna, Bangladesh [10], as shown in Fig. 3. The figure shows that the wind speed ranges from 2.64 to 4.33 m/s. The highest wind speed occurs in June. Monthly average solar radiation data is shown in Fig. 4. As used in HOMER software, the latitude and longitude of Khulna district are 22.5° N, 89.5° E, respectively. The annual average solar radiation is estimated to be 4.56 kWh/m^2/day [10].

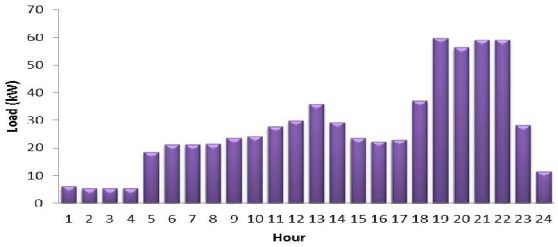


Fig.1. Load profile (daily).

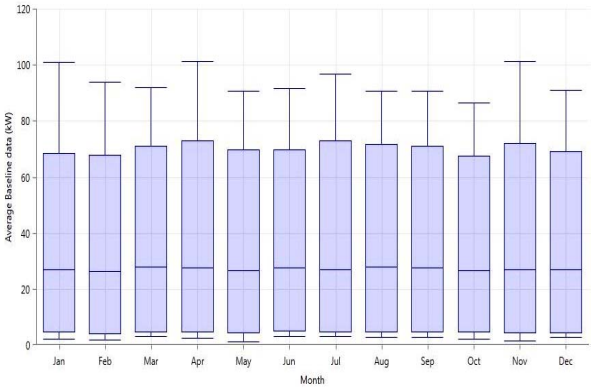


Fig. 2. Load for a complete year (monthly average).

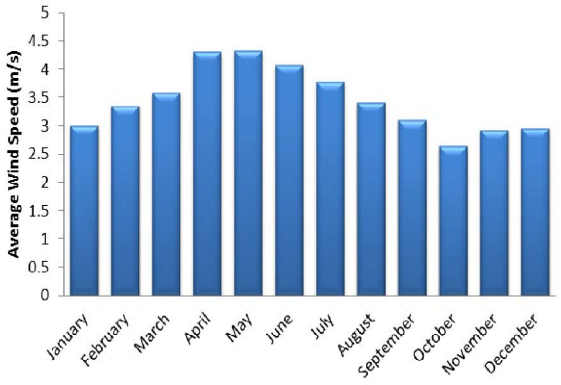


Fig. 3. Wind speed (monthly average).

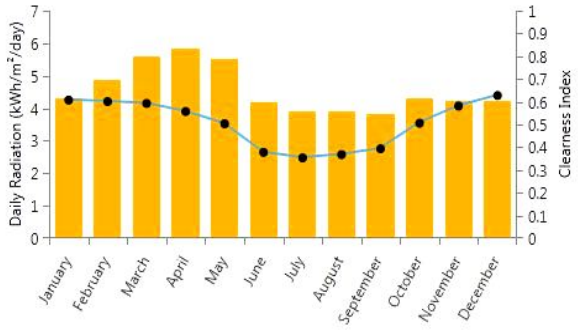


Fig. 4. Solar radiation and clearness index (monthly average).

III. Methodology

A. HOMER Software

National Renewable Energy Laboratory (NREL) in the United States (USA) developed HOMER software [10]. It is generally used for the design and analysis of hybrid power system. In this paper, electrical load, solar radiation and wind speed data, component details and costs are provided as input information to HOMER.

B. Cost Analysis Procedure by HOMER[10, 11, 12]

1) Net present cost (NPC): NPC indicates the installation cost and the operating cost of the system throughout its lifetime which is calculated as follows [12]:

$$NPC = TAC / CRF(i, Rpr_j)$$

where, TAC, CRF, i, and Rpr_j are the total annualized cost ($), capital recovery factor, interest rate in percentage, and project life time in year, respectively.

2) Total annualized cost: It is the sum of the annualized costs of every equipment of the power system including capital and operation and maintenance cost. It also includes replacement and fuel cost [12].

3) Capital recovery factor: It is a ratio which is used to calculate the present value of a series of equal annual cash flows [12].

$$CRF = \frac{i \times (1+i)^n}{(1+i)^{n-1}}$$

where, n and i represents the number of years and the annual real interest rate, respectively.

4) Annual real interest rate: It is a function of the nominal interest rate shown as [12]

$$i = \frac{i' - F}{1+F}$$

where, i, i' and F is the real interest rate, nominal interest rate, and annual inflation rate, respectively.

5) Cost of energy (COE): It is the average cost/kWh of useful electrical energy produced by the system.
The COE is calculated as follows [12]:

$$COE = \frac{TAC}{L_{prim,AC} + L_{prim,DC}}$$

where, $L_{prim,AC}$ and $L_{prim,DC}$ are the AC primary load and the DC primary load, respectively.

IV. Simulation Model

The components are chosen from HOMER to perform simulation. Figure 5 shows the grid connected hybrid power system design using HOMER which consists of PV array, wind generator, converter, load, battery and grid connection. Figure 6 shows the same deign but without grid connection.

To estimate the system performance under different situations, HOMER simulates the above two arrangements at the same area and same load based on different costs such as the estimated installation cost, operation and maintenance cost, replacement cost, interest, cost of energy.

The main components of the grid connected hybrid system are wind turbine, PV array, battery bank, and a power converter. For economic analysis, the following values are used.

A. Photovoltaic Array:

Name: Generic Flat Plate PV, manufactured by Generic. Its efficiency is about 13% and lifetime about 25 years. Initial installation cost of PV arrays is assumed as $3000 per kW and operation and maintenance cost practically zero.

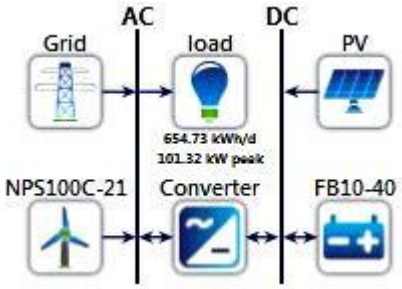


Fig. 5. The arrangement of hybrid power system (grid connected).

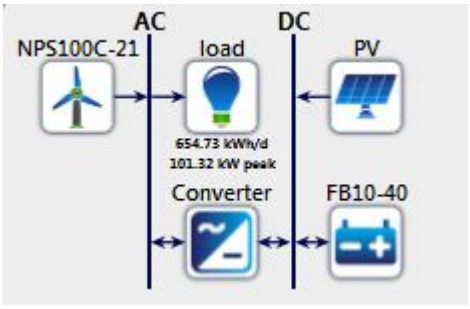


Fig. 6. The arrangement of hybrid power system (off-grid).

B. Wind Turbine:

For this study, three Northern Power NPS100C-21 100 kW wind turbines have been used [10]. Hub height is 23m, 30m, 37m, rotor 30.1m and its lifetime about 20 years. The capital and operation and maintenance costs are $32,000 and $300, respectively.

C. Power Converter:

Name: system converter, manufactured by Generic. Its efficiency is about 90% and lifetime about 20 years. In this paper, the capital cost for a 1 kW converter is $300. Replacement cost is $300 and operation and maintenance cost practically zero.

D. Battery Storage Bank:

The name of the battery bank is CELLCUBE®FB10-40, manufactured by Gildemeister, with a capacity 40 kWh, lifetime 876000 kWh. Its efficiency is about 64%. In grid-connected designs, a battery is only used during any fault on the grid to maintain the continuity of supply.

E. Grid:

In grid connected design, a grid is used as a backup power component or excess power absorber. The grid supplies power when there is not enough power from RES to meet the load demand and grid consumes power when excessive power is available.

Considering the availability of renewable energy sources in the considered area, wind and solar energy are used to design the hybrid power system. A battery energy storage is used to support the system in it's off grid mode.

V. Optimization Results

Figures 7 and 8 show the optimization results for the hybrid power model with and without grid connection, respectively. For grid connected system, as shown in Fig. 7, the minimum COE obtained from the result is $0.0995. In this scenario, the percentage of renewable energy contribution is 73%. In the proposed grid connected model, an optimum number of renewable energy sources is activated and supplies electricity to the load. From Fig. 8, it can be clearly seen that off grid power system for the same load is more expensive ($0.391) than grid connected system ($0.0995). The NPC for on-grid and off-grid design is $535,661 and $1.21M, respectively. From Fig. 9 it can be seen that the total yearly production of the proposed hybrid model is 436,186 kWh/yr and consumption by the load is 238,967 kWh/yr. The rest of the production is sold to the grid. The monthly average electricity production from different units using the proposed model is shown in Fig. 10.

Architecture									Cost				
					PV (kW)	NPS100C-21	FB10-40	Grid (kW)	Converter (kW)	COE ($)	NPC ($)	Operating cost ($)	Initial capital ($)	Ren Frac (%)
					150	3		999,999	180	$0.0995	$535,661	-$4,977	$600,000	73
					150	3	2	999,999	180	$0.101	$542,572	-$4,906	$606,000	73
					200			999,999	180	$0.134	$662,548	$661.25	$654,000	62
					200		2	999,999	180	$0.135	$669,536	$737.62	$660,000	62

Fig. 7. Screenshot of simulation for finding optimal design (grid connected).

Architecture						Cost				
PV (kW)	NPS100C-21	FB10-40	Converter (kW)	COE ($)	NPC ($)	Operating cost ($)	Initial capital ($)	Ren Frac (%)		
300	3	40	180	$0.391	$1.21M	$2,773	$1.17M	100		

Fig. 8. Screenshot of simulation result of off grid power system (PV/Wind/Battery).

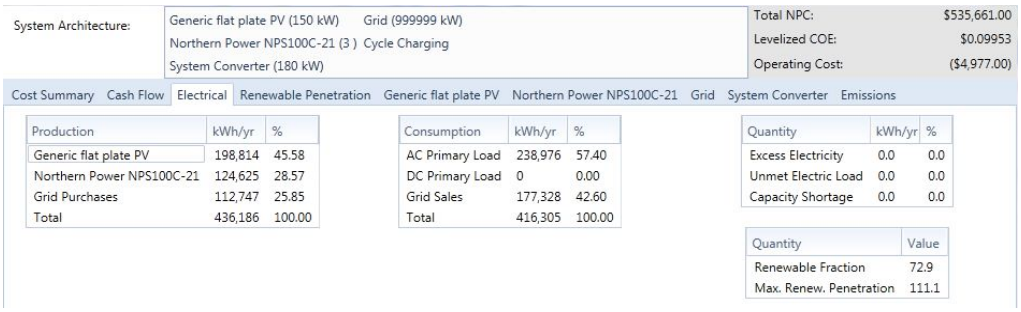


Fig. 9. Snapshot of the production and consumption scenario.

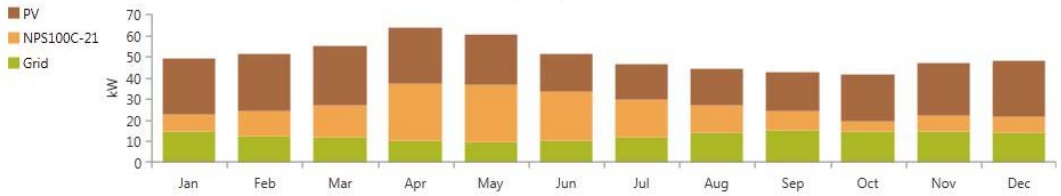


Fig. 10. Monthly average electricity production using the proposed model.

VI. Conclusion

This paper presents a comparative analysis between an off-grid and a grid connected hybrid power system in a specific area in Bangladesh. The optimization result shows that the grid connected hybrid (PV/wind) system is more efficient and economic compared to the traditional hybrid system (PV/Wind/Battery) for the same load. From the simulation result, it is also investigated that Net Present Cost of the proposed model is less than off-grid model. Although off-grid hybrid system uses 100% renewable energy but it needs an extra-large battery bank for storage of electricity. On the other hand, excess output power of off-grid system goes unused. A grid connected hybrid system does not require an extra battery bank in normal operating condition and excess power that is produced by renewable energy are added to the grid. Therefore, it can be said that the proposed grid connected hybrid power system is most suitable and cost effective as it offers several benefits.

References

[1] http://www.bpdb.gov.bd/bpdb/index.php

[2] U. Sureshkumar, P. S. Manoharan, and A.P.S. Ramalakshmi, "Economic cost analysis of hybrid renewable energy system using HOMER," International Conference on Advances in Engineering, Science And Management, pp. 94-99, March 30-31, 2012.

[3] Ahmad Rohani, Kazem Mazlumi, and Hossein Kord, "Modeling of a hybrid power system for economic analysis and environmental impact in HOMER," 18th Iranian conference on Electrical Engineering (ICEE), , pp. 819-823, 11-13 May 2010.

[4] W. D. Kellogg, M. H. Nehrir, V. Gerez, and G. V. Venkataramanan, "Generation unit sizing and cost analysis for stand-alone wind, photovoltaic and hybrid wind/pv systems," IEEE Trans. on Energy Conversion, vol. 13, pp.70-75, 1998.

[5] F. Giraud, and Z. M. Salameh, "Steady-state performance of a grid connected rooftop hybrid wind-photovoltaic power system with battery storage," IEEE Trans. on Energy Conversion, vol. 16, no.1, pp. 1-7, 2001.

[6] R. Ramakumar, I. Abouzahr, and K. Ashenayi, "A knowledge-based approach to the design of integrated renewable energy systems," IEEE Trans. on Energy Conversion, , vol. 7, no.4, pp.648-659, 1992

[7] R. Chedid, H. Akiki, and S. Rahman, "A Decision Support Technique for the Design of Hybrid Solar-Wind Power Systems," IEEE Trans. on Energy Conversion, vol. 13, no.l, pp.76-83, 1998.

[8] W. O. Kellogg, M. H. Nehrir, G. Venkataramanan and V. Gerez, "Generation unit sizing and cost analysis for stand-alone wind, photovoltaic, and hybrid wind pv systems," IEEE Trans. on Energy Conversion, vol. 3, no.l, pp.70-75, 1998.

[9] B. E. Turkay and A. Y. Telli, "Economic analysis of standalone and grid connected hybrid energy systems," International Conference on Electrical and Electronics Engineering, pp.l-34-1-39, 5-8 Nov., 2009.

[10] www.nrel.gov/homer

[11] Getting Started Guide for HOMER Version 3.3.1; (https://www.homerenergy. com).

[12] H. Rezzouk and A. Mellit, "Feasibility study and sensitivity analysis of a stand-alone photovoltaic–diesel–battery hybrid energy system in the north of Algeria," Renewable and Sustainable Energy Reviews, vol. 43, pp. 1134–1150, 2015.